

Identify and analyze Hot Topics Related to Israel and Jews

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ABSTRACT

This study analyzes discourse across four Reddit communities (r/Israel, r/Judaism, r/Jewish, r/ReformJews) from October 2023 to November 2024, combining topic modeling (LDA), Named Entity Recognition (NER), sentiment and temporal analysis, and network analysis to uncover key themes, influential users, and shifts in sentiment during critical events. Topic modeling identified eight recurring themes ranging from political conflict to religious identity. NER map prominent entities, exposing how geopolitical events, historical memory, and religious holidays shape discussions. Sentiment analysis showed that posting spikes coincide with sentiment drops, especially following crises such as October 7 and the Iranian missile attack. Network analysis revealed a small group of users driving discourse, with political discussions dominated by highly polarizing figures, while cultural topics attracted more neutral voices. Case studies of peak activity dates demonstrate how external crises trigger internal identity discourse, blending real-time news sharing with cultural reflection. Together, these insights highlight how online Jewish and Israeli discourse intertwines personal identity, collective memory, and global events.

1 INTRODUCTION

Online communities play a crucial role in shaping public discourse around identity, politics, and global events. On platforms like Reddit, communities dedicated to specific topics—subreddits—become spaces where users negotiate personal and collective identities, respond to real-world crises, and shape narratives through discussion [2, 8]. In particular, discussions surrounding Israel and Judaism are deeply intertwined with historical memory, political polarization, and religious identity [3, 6]. Systematically mapping these conversations provides valuable insight into how global events, such as conflicts or antisemitic incidents, ripple into online spaces, affecting both sentiment and thematic focus. This study applies a multi-method approach to analyze discussions across four relevant subreddits, combining topic modeling, NER, sentiment analysis, and network analysis. This integration captures both the content and emotional tone of conversations, while also identifying the key users shaping discourse and tracing how offline events influence online dialogue.

2 BACKGROUND

The study of online discourse, particularly in contexts of political conflict and identity negotiation, has received growing attention in recent years. In particular, studies focusing on discussions about Israel and Jewish identity have demonstrated how online discourse intertwines religious, political, and historical frames, often amplifying polarization around contested events [3, 6]. Prior works have also applied topic modeling to map discourse landscapes [10], while others leveraged NER to track key actors and geopolitical focus over time [9]. Sentiment analysis has been used extensively to capture the emotional dynamics triggered by political violence and antisemitic incidents, showing that spikes in negativity tend to coincide with crisis moments [7]. However, relatively few studies have combined these methods into a unified pipeline that captures not only content and sentiment, but also the networked interaction between key users, themes, and real-world events over time. By integrating topic modeling, NER, sentiment analysis, and leadership network analysis, this study builds on these foundations to provide a comprehensive, event-sensitive view of discourse evolution within Reddit’s Israel- and Judaism-focused communities.

3 DATASET

3.1 Data Collection

The data used in this study was collected from Reddit, a popular online platform that hosts a wide range of communities (subreddits) dedicated to various topics. The data spans the period from October 2023 to November 2024.

The dataset includes both posts and comments from four subreddits that focus on topics related to Israel and Judaism: r/Israel, r/Judaism, r/Jewish, and r/ReformJews. These subreddits were selected because they represent diverse discussions surrounding Jewish identity, religion, culture, and current events related to Israel.

Table 1 provides a summary of the number of posts and comments collected from each subreddit. In total, the dataset contains 76,271 posts and 1,495,139 comments, resulting in 1,571,410 data points overall.

Table 1: Number of posts and comments from different subreddits.

Subreddit	Posts	Comments	Total
Israel	35,711	674,314	710,025
Judaism	20,149	390,386	410,535
Jewish	19,797	423,701	443,498
ReformJews	614	6,738	7,352
Total	76,271	1,495,139	1,571,410

4 METHODOLOGY

4.1 Topic Modeling (LDA)

To uncover the main themes discussed across the four subreddits (r/Israel, r/Jewish, r/Judaism, r/ReformJews), we applied **Latent Dirichlet Allocation (LDA)** — a probabilistic model that identifies hidden topics within a collection of texts. LDA assumes each document contains a mixture of topics, and each topic is represented by a set of words with different weights.

The topic modeling process consisted of several key steps. **Preprocessing** involved cleaning posts and comments by removing stopwords, converting to lowercase, tokenizing, and lemmatizing. **Model Training** was performed exclusively on posts, under the assumption that posts capture the primary themes, which are later reflected in comments. To determine the **Optimal Number of Topics**, we computed coherence scores (Appendix Figure 3) to balance semantic clarity and model stability. Following training, **Comment Classification** assigned each comment to one of the identified topics based on its word distribution. Each topic was then described through **Topic Representation**, consisting of top words ranked by both frequency (how often a word appeared) and relevance (a combination of frequency and exclusivity calculated using **pyLDAvis**). Finally, for **Topic Labeling**, the top words and their weights were fed into **GPT**, which generated clear and interpretable topic labels.

4.2 Named Entity Recognition (NER)

To enrich our understanding of the discourse within the analyzed subreddits, we applied NER to detect and classify key entities appearing in the discussions. We utilized **spaCy**, a widely used NLP library, employing a model optimized for social media and informal text. The extracted entities were categorized into five main types: **GPE**(locations); **PERSON**; **ORG** (organizations); **DATE**; and **EVENT**.

Following entity extraction, we constructed **entity co-occurrence networks** for each topic, as identified by the LDA models. These networks capture pairwise co-occurrence frequencies, reflecting which entities tend to appear together within the same post or comment. This step allowed us to detect clusters of actors and locations, revealing underlying relational structures embedded in user discourse.

4.3 Temporal and Sentiment Analysis & BERTopic

To investigate how sentiment and thematic focus evolve during critical events, we applied a combined approach of sentiment analysis, volume tracking, and event-specific topic modeling using BERTopic.

We first constructed a daily time series for each subreddit, capturing both post volume and average sentiment. Sentiment for each post was computed using **VADER** [?], a lexicon-based tool optimized for informal online text. Each post was classified as positive (score > 0.2), neutral (-0.2 ≤

score ≤ 0.2), or negative (score < -0.2). To smooth short-term fluctuations, a 7-day moving average was applied to both sentiment and volume trends.

To detect key moments, we identified extreme dates based on two criteria: (1) days with the highest and lowest average sentiment, and (2) days with peak posting volume. For each extreme date, we analyzed a 3-day window covering the day before, the event day, and the day after, to capture the immediate lead-up and follow-up discourse.

For posts within each window, we applied BERTopic [4], a transformer-based topic modeling approach designed for short and noisy texts. BERTopic identified the dominant topics, providing top terms, topic distributions, and visualizations. This allowed us to trace how the thematic focus shifted in response to these extreme events.

4.4 Network Analysis & Leadership

In online communities, a small group of highly active users often shapes the discourse, drives engagement, and influences sentiment. To identify these influential contributors, we developed a leadership score that integrates multiple aspects of user activity and visibility. This approach was inspired by prior work on leadership and influence in social networks, including LeaderRank [5], which evaluates user importance based on network flow, and Berger and Milkman [1], who explored how content virality is driven by both linguistic properties and social transmission dynamics.

To adapt these principles to Reddit’s unique structure, we defined leadership as a weighted combination of user activity and engagement metrics:

$$\text{Leadership} = 0.3 \cdot \text{Posts} + 0.3 \cdot \text{Comments} + 0.2 \cdot \text{Upvotes} + 0.2 \cdot \text{Replies} \quad (1)$$

This formula balances content creation (posts and comments) with indicators of community impact (total upvotes received and replies to their content), ensuring leadership reflects both the volume of participation and the extent to which others engage with their contributions.

To further explore how these leading users interact with each other, we constructed a directed reply network, where nodes represent top authors (ranked by leadership score) and edges indicate that one author replied to another. We computed standard centrality measures: degree centrality, reflecting direct connectivity; betweenness centrality, capturing users who act as bridges between otherwise disconnected groups; and clustering coefficient, showing the extent to which each user’s connections form tightly-knit clusters.

In addition, we computed a **polarization score** for each top author, measuring how often they employ "us versus them" language, such as using pronouns like "we" and "they", which serves as a proxy for divisive, group-based framing. This measure allows us to capture the extent to which each leader frames discussions in polarized, conflictual terms. Alongside this, we computed the **average sentiment** of each user, providing a complementary view of the emotional tone of their contributions.

5 EVALUATION

5.1 Topic Modeling (LDA)

As part of the analysis, we applied topic modeling to identify the main topics discussed on the subreddits. As shown in Appendix Figure 6, we trained multiple LDA models with different numbers of topics and evaluated their semantic coherence. We selected $k = 8$ to maintain clear interpretability and to avoid over-fragmentation of themes.

As shown in Appendix Table 7 and Appendix Table 8, we present the top 15 words for each topic based on two measures: frequency and relevance. Frequency highlights the most common terms within each topic, while relevance emphasizes words that are both frequent and distinctive to the topic (i.e., adjusted for exclusivity). Together, these tables provide a richer understanding of how each topic is characterized. To further illustrate the essence of each topic, Figure 1 presents word clouds, which offer a visual snapshot of the characteristic terms that define each topic. Combining these numerical (tables) and visual (word clouds) views, allows us to interpret each topic in context and highlight its core meaning.



Figure 1: Word clouds for each topic identified by the LDA model.

Topic 0: Emotional Reactions to Conflict & Politics — This topic captures emotional expressions within the broader political reality. Words like *feel*, *friend*, and *hate* highlight emotional and social reactions, while terms like *Israel* and *Gaza* link it back to geopolitical events.

Topic 1: Religious Identity — Focuses on religious identity, conversion, and differences within Judaism. Words like *convert*, *reform*, *orthodox*, and *religion* highlight discussions about religious and personal journeys into Judaism.

Topic 2: Culture & Antisemitism — Covers Jewish customs like *kosher* and *shabbat*, alongside discussions about *antisemitism* and *protests* on university campuses, showing the intersection of culture and external hostility.

Topic 3: Settlements & Nationalism — Centers on land disputes, settler policies, and national identity, with terms like *arab*, *land*, *nation*, and *settler* dominating the discussion.

Topic 4: Terrorism & Violence — Focuses on violent incidents, including *hostage* situations and *terrorist* attacks, dominated by terms like *hamas*, *kill*, and *massacre*.

Topic 5: Holidays & Traditions — Highlights cultural and religious practices, with terms like *passover*, *music*, and

shabbat, emphasizing celebrations and traditions.

Topic 6: Global Politics & Leaders — Dominated by terms like *iran*, *biden*, *netanyahu*, and *minister*, reflecting both domestic Israeli leadership and its intersection with international politics.

Topic 7: Campus Protests & Free Speech — Focused on *student*, *campus*, *ban*, and *university*, this topic involves discussions around academic activism, freedom of expression, and protest movements.

While each topic presents a distinct thematic core, the relationships between the topics themselves offer additional insights into how discussions cluster and overlap across the corpus. Therefore, we also examined an Intertopic Distance Map generated by pyLDavis. (Appendix Figure ??) visualizes the semantic proximity between topics, with closer circles representing thematically similar topics. In particular, Topics 6 and 7 overlap heavily, indicating that discussions about global political leaders (Topic 6) are closely tied to debates around campus protests and free speech (Topic 7), reflecting how international politics influences discourse within academic environments. Furthermore, Topics 7, 8, and 2 form a connected cluster, highlighting the intertwined nature of religious identity (Topic 8), cultural and antisemitism discourse (Topic 2), and campus activism (Topic 7). Topics 1 and 4 also appear close together, indicating some thematic overlap between religious identity (Topic 1) and terrorism and violence (Topic 4), potentially due to religious framing in discussions of violence and extremism. In contrast, Topic 3 (Settlements & Nationalism) stands apart from all other topics, suggesting that it forms a relatively isolated area of discussion. Topic 5 (Holidays & Traditions) is centrally positioned, potentially reflecting its relevance across different conversations, from religious identity to cultural customs and even political contexts.

Beyond semantic proximity, we also examined how these topics are distributed across different subreddits, shedding light on the unique character of each community. The topic distributions across subreddits (Appendix Figure 10) highlight clear differences between the communities. In **r/Israel** and **r/Jewish**, the leading topic is **Global Politics & Leaders** (Topic 6), reflecting their focus on news, geopolitical developments, and political discourse relevant to Israel and the Jewish diaspora. In contrast, **r/Judaism** and **r/ReformJews** are dominated by **Campus Protests & Free Speech** (Topic 7), aligning with their emphasis on religious identity, cultural inclusion, and academic activism. These patterns align well with the distinct character of each subreddit, strengthening the validity of the topic modeling results.

5.2 Named Entity Recognition (NER)

NER enables the identification of key entities discussed in the subreddits, linking abstract topics to concrete actors, locations, and events. This analysis adds essential historical,



Figure 2: Geographic Distribution of the Mentioned GPE Across All Topics

political, and cultural context, highlighting who and what dominates the discourse.

Table 2 presents the leading entities for each topic, broken down by entity type.

Beyond the expected centrality of Israel, the discussions are heavily shaped by the presence of other geopolitical entities such as Iran, Gaza, Russia, and Lebanon, highlighting the strong focus on Middle Eastern geopolitics. This global scope is visually reflected in Figure 2, which highlights the top mentioned locations across all topics and their geographic spread.

Political leaders including Netanyahu and Biden emerge as dominant figures, reflecting the close connection between leadership and political discourse. Militant and political organizations like Hamas and Hezbollah are highly visible, particularly in conflict-related topics, while religious and cultural organizations such as Chabad appear mainly in religious discussions. Historical and religious dates, including Yom Kippur, play a significant role, anchoring conversations in shared memory and symbolic events. Additionally, historical events like World War II frequently appear, demonstrating how users actively link past conflicts to contemporary issues.

To understand how entities relate to one another within a given topic, we constructed co-occurrence networks, as illustrated in Figure 3, which shows how in Topic 0, Israel serves as the central hub around which military actors like Hamas, Hezbollah, and Iran cluster tightly reflecting the dominance of conflict-related discourse. Notably, Ukraine emerges as a connected node, indicating that the war in Ukraine is frequently mentioned alongside discussions about Israel, possibly in the context of global instability, international alliances, or historical parallels between conflicts.

In addition to mapping overall entity networks, we conducted focused analyses on specific entities to uncover how they distribute across topics and which other entities co-occur with them most frequently. A notable case is World War II, whose co-occurrence network (Appendix Figure 11) reveals how historical memory shapes the discourse. WWII appears most often alongside Israel, reflecting the central role of the Holocaust in modern Jewish and Israeli identity. Interestingly, Hamas and Gaza also frequently co-occur with WWII, indicating that users actively frame the current conflict through historical analogies. The mentions of Germany,

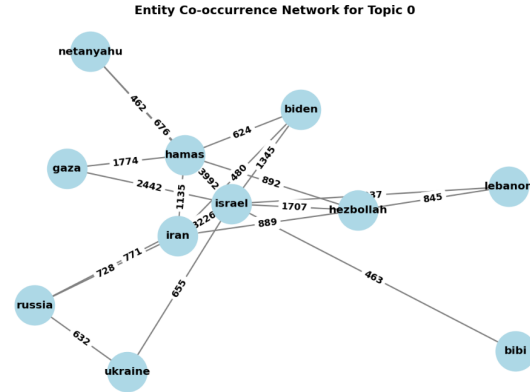


Figure 3: Entity Co-occurrence Network for Topic 0



Figure 4: Daily Post Count and Average Sentiment Over Time

Poland, and Russia connect to the broader geographical context of the war, while the recurring appearance of terms like today and America highlights how users draw explicit parallels between historical and contemporary events.

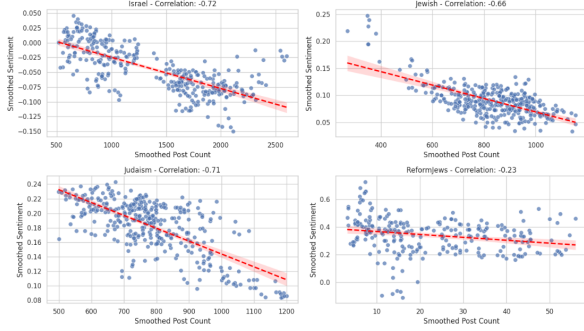
5.3 Temporal and Sentiment Analysis & BERTopic

Sentiment analysis across topics (Appendix Figure 12) highlights the emotional intensity of certain themes. Topics linked to "Terrorism & Violence" and "Settlements & Nationalism" show particularly high negativity, while topics focused on "Holidays & Traditions" and "Religious Identity" display more positive sentiment.

To capture the temporal evolution of sentiment and volume, we plotted daily post count and average sentiment (Figure 4). These trends reveal that spikes in posting volume frequently coincide with sharp drops in sentiment, indicating that major crises and political events trigger both heightened activity and more negative discourse. This pattern is especially clear in r/Israel, where events such as the October 7 anniversary and the Iranian missile attack directly correspond with significant sentiment drops. The correlation analysis (Figure 5) confirms this inverse relationship, with strong negative correlations between post volume and sentiment across all subreddits. This reinforces the finding that surges in activity typically reflect reactions to negative events, driving more emotionally charged discussions.

Table 2: Top 3 Named Entities per Topic and Entity Type

Topic	GPE	PERSON	ORG	DATE	EVENT
0	israel, iran, gaza	biden, netanyahu, bibi	hamas, hezbollah, idf	today, october, oct	lebanon war, olympics, yom kippur war
1	israel, tel aviv, jerusalem	magen david, david, torah	hamas, idf, amazon	today, friday, year	olympics, hanukkah, shabbat shalom
2	israel, palestine, germany	modmail, hitler, biden	hamas, adl, wikipedia	today, october, oct	olympics, wwii, auschwitz
3	israel, gaza, palestine	netanyahu, bibi, biden	hamas, idf, hezbollah	october 7th, october, oct	geneva convention, wwii, olympics
4	israel, palestine, gaza	netanyahu, jordan, biden	hamas, hezbollah, plo	today, 1948, 1967	wwii, geneva convention, olympics
5	israel, jerusalem, egypt	torah, asher, god	hamas, youtube, ord	today, yom kippur, weekly	sukkot, hanukkah, olympics
6	israel, gaza, palestine	biden, netanyahu, bibi	hamas, hezbollah, idf	today, october 7th, october	olympics, wwii, vietnam war
7	israel, poland, germany	jesus, torah, kabbalah	hamas, islam, nonorthodox	today, year, year ago	wwii, hanukkah, jewish day

**Figure 5: Correlation Between Post Volume and Sentiment Across Subreddits****Table 3: Extreme Dates Identified Per Subreddit**

Subreddit	Min Sentiment	Max Sentiment	Min Posts	Max Posts
Israel	2024-10-07	2024-11-20	2024-10-07	2024-04-14
Jewish	2024-03-05	2023-10-04	2024-02-11	2024-05-01
Judaism	2024-05-11	2024-07-20	2024-07-20	2023-11-12
ReformJews	2024-07-07	2024-09-26	2024-07-07	2023-10-26

Table 3 summarizes the dates with the highest and lowest sentiment, as well as peak posting days for each subreddit. Peak posting days often correspond to geopolitical crises, antisemitic incidents, or religious holidays.

Among the identified peak activity dates, October 7, 2024, emerged as a critical point in r/Israel, marking the anniversary of the October 7, 2023 Hamas attack. This date triggered a significant spike in posting volume, as users engaged in intense discussions about the attack’s legacy, its political and security aftermath, and broader geopolitical ramifications. The BERTopic analysis (Appendix Figure 13) reveals that the discourse was dominated by crisis-related themes, including references to hostages, military actions, and international responses. Many posts revisited the initial attack, debated Israel’s military strategy, and highlighted personal experiences of fear and trauma. This reinforces the role of Reddit not only as a platform for real-time news sharing but also as a space for collective memory construction, where users revisit and reframe past events in light of present tensions.

5.4 Network Analysis & Leadership

The network analysis revealed that influence across the subreddits is concentrated among a small group of highly active users. Table 4 lists the top-ranked authors based on their leadership scores, with **OkBuyer1271**, **anon755qubwe**, and **welltechnically7** standing out due to their extensive posting, commenting, and engagement from others. These users do not influence all topics equally. As shown in the topic dominance heatmap (Appendix Figure 14),

Table 4: Top 5 Authors by Leadership Score

Author	Leadership Score	Avg Sentiment	Avg Polarization
OkBuyer1271	2317.34	-0.002	0.0136
anon755qubwe	1833.06	-0.112	0.0210
welltechnically7	1669.23	0.074	0.0165
ummbacon	1507.67	0.102	0.0184
Middle_Ad_8052	1296.75	-0.189	0.0141

anon755qubwe and **Middle_Ad_8052** are most dominant in political and conflict-oriented discussions (Topics 0 and 3), aligning with their higher polarization scores. In contrast, **ummbacon** and **welltechnically7** focus more on religious and cultural topics (Topics 1 and 5), where polarization is lower, and sentiment tends to be more neutral or positive.

The full reply network of the top 20 authors appears in Appendix Figure 15, illustrating how these leading users interact with one another. The network highlights a clear core-periphery structure, where highly central users such as **welltechnically7** and **anon755qubwe** act as conversational hubs, while others maintain more peripheral roles.

A strong link emerges between polarization and sentiment. Highly polarizing users—who frequently use "we" and "they" language—dominate political and conflict discussions, often with more negative sentiment. In contrast, less polarizing users, who focus on religious and cultural topics, exhibit more balanced or positive sentiment.

6 CONCLUSIONS AND FUTURE WORK

This study provides a multi-dimensional view of discourse across Reddit communities focused on Israel and Judaism, revealing how online discussions blend personal identity, historical memory, and global events. By combining topic modeling, entity recognition, sentiment tracking, and network analysis, we exposed both the dominant themes and the key users shaping these conversations. The findings highlight how political conflict triggers polarized and emotionally charged discourse, while cultural and religious discussions foster more balanced participation. Our temporal analysis further shows that real-world crises directly impact online sentiment and thematic focus, illustrating the dynamic interplay between online discourse and offline events. Future work could extend this analysis by extracting relevant discussions from subreddits not explicitly focused on Israel, incorporating data from additional platforms such as Twitter and TikTok, and conducting longitudinal sentiment analysis over a longer period to capture evolving trends.

7 ACKNOWLEDGMENT

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GitHub Repository:

https://github.com/edenluzon5/BIG_DATA_PROJECT/

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A APPENDIX

A.1 Topic Modeling (LDA)

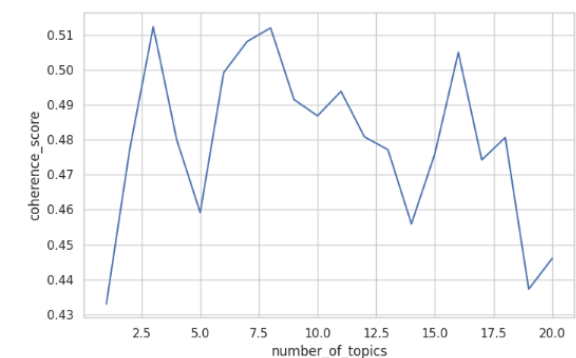


Figure 6: Coherence scores for different numbers of topics.

	Topic 0	Weights 0	Topic 1	Weights 1	Topic 2	Weights 2	Topic 3	Weights 3	Topic 4	Weights 4	Topic 5	Weights 5	Topic 6	Weights 6	Topic 7	Weights 7
0	israel	0.048365	get	0.012776	jewish	0.031020	hamas	0.033353	israel	0.035413	torah	0.010311	jewish	0.021632	jewish	0.087120
1	war	0.016757	know	0.010906	antisemitism	0.019679	gaza	0.018545	jewish	0.012505	one	0.007884	people	0.020991	convert	0.014172
2	iran	0.012259	would	0.010740	student	0.016938	kill	0.013495	state	0.011834	book	0.007510	like	0.014985	reform	0.009689
3	israeli	0.011148	want	0.010572	antisemitic	0.012507	israeli	0.013016	palestinian	0.011267	song	0.007069	feel	0.013893	would	0.009220
4	gaza	0.010794	like	0.009304	university	0.011162	hostage	0.012940	country	0.010899	day	0.007009	say	0.012771	community	0.008871
5	say	0.009658	work	0.008451	post	0.011079	october	0.011345	israeli	0.010362	say	0.006918	think	0.011621	family	0.008673
6	hamas	0.009548	look	0.008344	protest	0.010035	attack	0.011200	arab	0.008938	god	0.006668	know	0.011221	know	0.008215
7	hezbollah	0.008936	make	0.008148	campus	0.008820	israel	0.011117	land	0.007554	music	0.006419	get	0.009777	rabbi	0.007215
8	army	0.008498	go	0.007922	zionism	0.008141	terrorist	0.011094	people	0.006890	read	0.006270	see	0.009304	like	0.007086
9	idf	0.007824	anyone	0.007887	holocaust	0.006403	palestinian	0.010864	world	0.006688	prayer	0.006101	make	0.008643	question	0.006492
10	attack	0.007789	year	0.007750	flag	0.006115	civilian	0.008140	would	0.005723	make	0.006008	israel	0.008521	learn	0.006274
11	netanyahu	0.007254	shabbat	0.007080	nazi	0.005834	idf	0.007882	zionism	0.005052	year	0.005960	even	0.008477	want	0.006196
12	lebanon	0.005890	time	0.006988	call	0.005765	child	0.007432	palestine	0.005048	story	0.005591	want	0.008453	religious	0.005767
13	iranian	0.005448	find	0.006664	college	0.005376	woman	0.006825	history	0.004997	first	0.005326	dont	0.007925	orthodox	0.005588
14	ceasefire	0.005091	one	0.006201	group	0.005335	death	0.006747	muslim	0.004615	rabbi	0.005212	friend	0.007248	religion	0.005534

Figure 7: Top 15 Frequent Words for Each Topic

	Topic 0	Relevance 0	Topic 1	Relevance 1	Topic 2	Relevance 2	Topic 3	Relevance 3	Topic 4	Relevance 4	Topic 5	Relevance 5	Topic 6	Relevance 6	Topic 7	Relevance 7
0	people	0.216470	convert	0.884640	wear	0.737555	arab	1.078820	hamas	1.548570	song	1.543005	iran	1.771965	student	1.718375
1	feel	0.144195	reform	0.827680	kosher	0.699130	land	1.047570	terrorist	1.508115	music	1.488765	hezbollah	1.740940	ban	1.707715
2	hanukkah	0.024940	jewish	0.714700	shul	0.667635	settler	0.926385	hostage	1.467195	christmas	1.476635	netanyahu	1.713165	campus	1.688500
3	see	0.023285	orthodox	0.712565	shabbat	0.633320	nation	0.924905	civilian	1.432930	passover	1.470695	biden	1.649645	university	1.617220
4	bad	0.015275	religion	0.659190	buy	0.631485	state	0.896270	kill	1.416630	megathread	1.460205	minister	1.643620	professor	1.613330
5	friend	0.007760	ashkenazi	0.632965	email	0.558120	west	0.836535	soldier	1.362610	yom	1.433380	iranian	1.636290	columbia	1.601640
6	dont	-0.003170	religious	0.629910	travel	0.556290	menorah	0.832690	massacre	1.362420	kippur	1.427790	lebanon	1.604980	protest	1.561140
7	happen	-0.014935	raise	0.623935	trip	0.545350	country	0.823405	hospital	1.352535	sing	1.411255	court	1.593455	encampment	1.558550
8	hate	-0.018240	mother	0.620330	kippah	0.532125	indigenous	0.808155	murder	1.335770	seder	1.403610	bibi	1.583585	san	1.538030
9	talk	-0.018665	catholic	0.606520	necklace	0.531735	east	0.805065	kidnap	1.333775	rosh	1.390210	prime	1.537755	flag	1.522630
10	even	-0.029100	practice	0.590145	service	0.531190	mezuzah	0.800600	release	1.330450	challah	1.380155	election	1.523295	mod	1.517985
11	think	-0.029475	christian	0.573450	shop	0.520410	colonial	0.799790	rap	1.304605	candle	1.376380	chief	1.490245	zealand	1.512125
12	fire	-0.031675	grandmother	0.570280	tattoo	0.517345	1948	0.797330	death	1.304405	movie	1.364365	trump	1.479085	wikipedia	1.504725
13	thing	-0.033830	christianity	0.563925	suggestion	0.514965	bank	0.788590	gaza	1.288790	purim	1.355830	warrant	1.476130	poster	1.500735
14	fuck	-0.051080	father	0.561930	work	0.503995	settlement	0.779015	footage	1.273355	podcast	1.353245	icj	1.472745	nazi	1.499685

Figure 8: Top 15 Words by Relevance for Each Topic

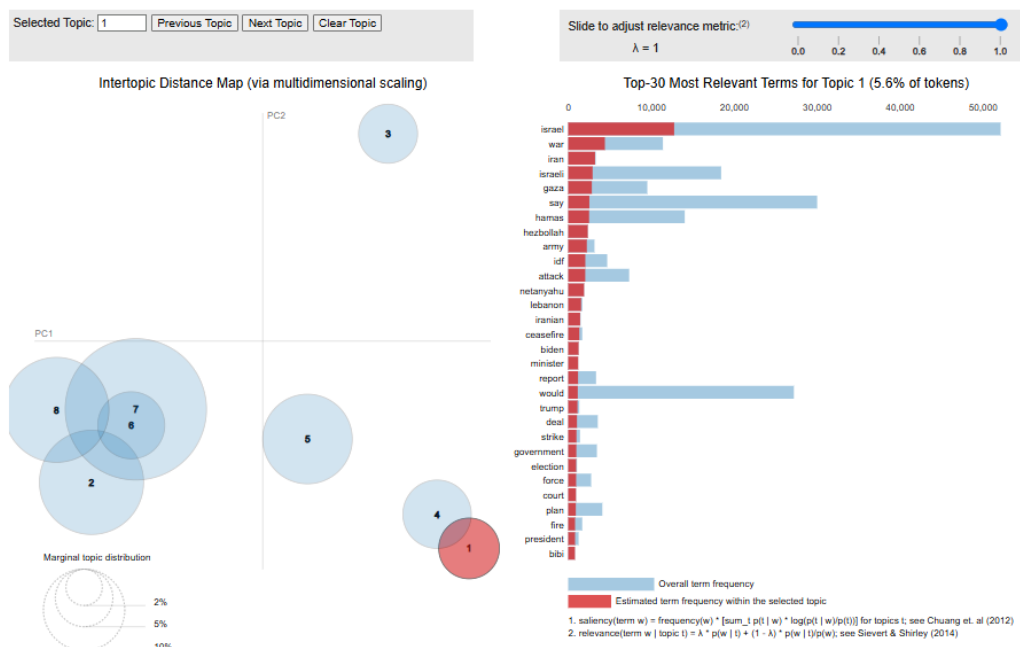


Figure 9: Intertopic Distance Map (pyLDavis Visualization).

Note: The topic numbering in this figure starts from 1 (i.e., Topic 1 in this figure corresponds to Topic 0 in the main analysis, Topic 2 corresponds to Topic 1, and so forth).

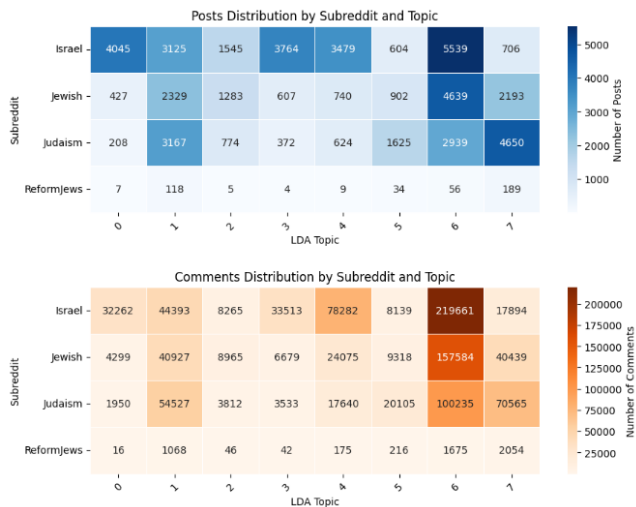


Figure 10: Posts and Comments Distribution by Subreddit and Topic

A.2 Named Entity Recognition (NER)

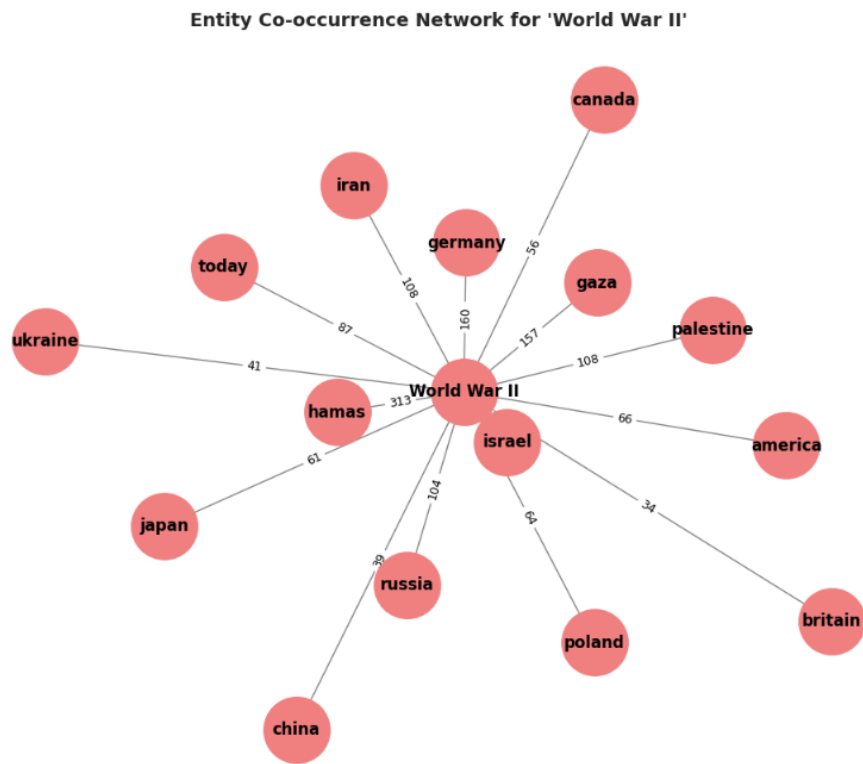


Figure 11: Entity Co-occurrence Network for World War II

A.3 Temporal and Sentiment Analysis & BERTopic

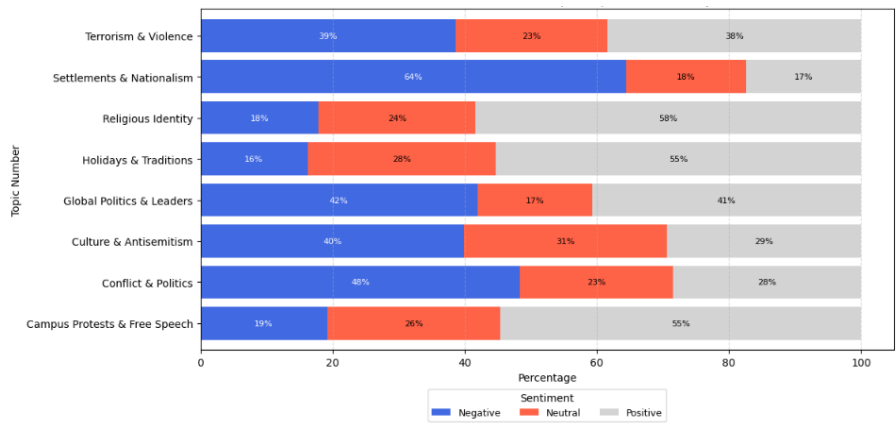


Figure 12: Sentiment Distribution Across Subreddits

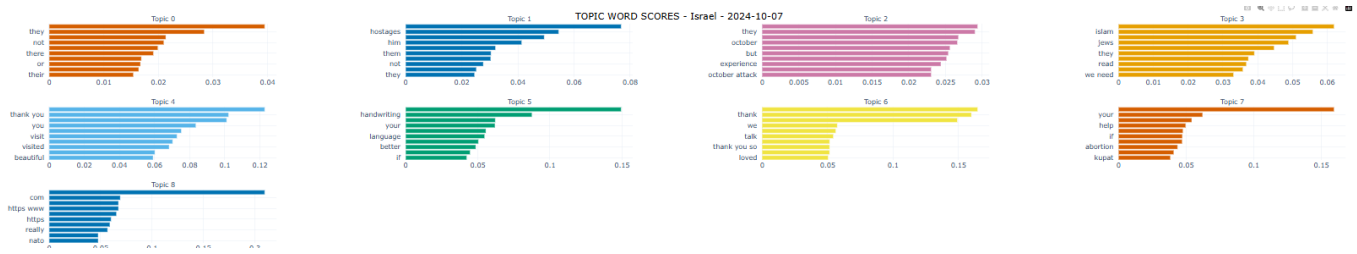


Figure 13: BERTopic Analysis for October 7, 2024 — r/Israel

A.4 Network Analysis & Leadership Detection

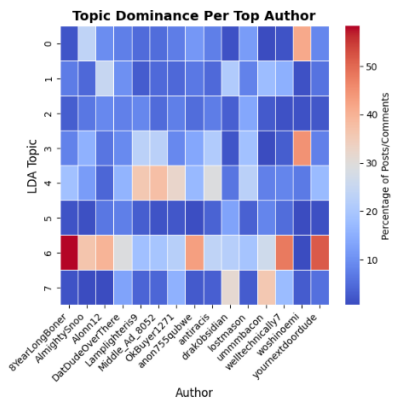


Figure 14: Topic Dominance Among Top 20 Authors



Figure 15: Reply Network of Top 20 Authors by Leadership Score